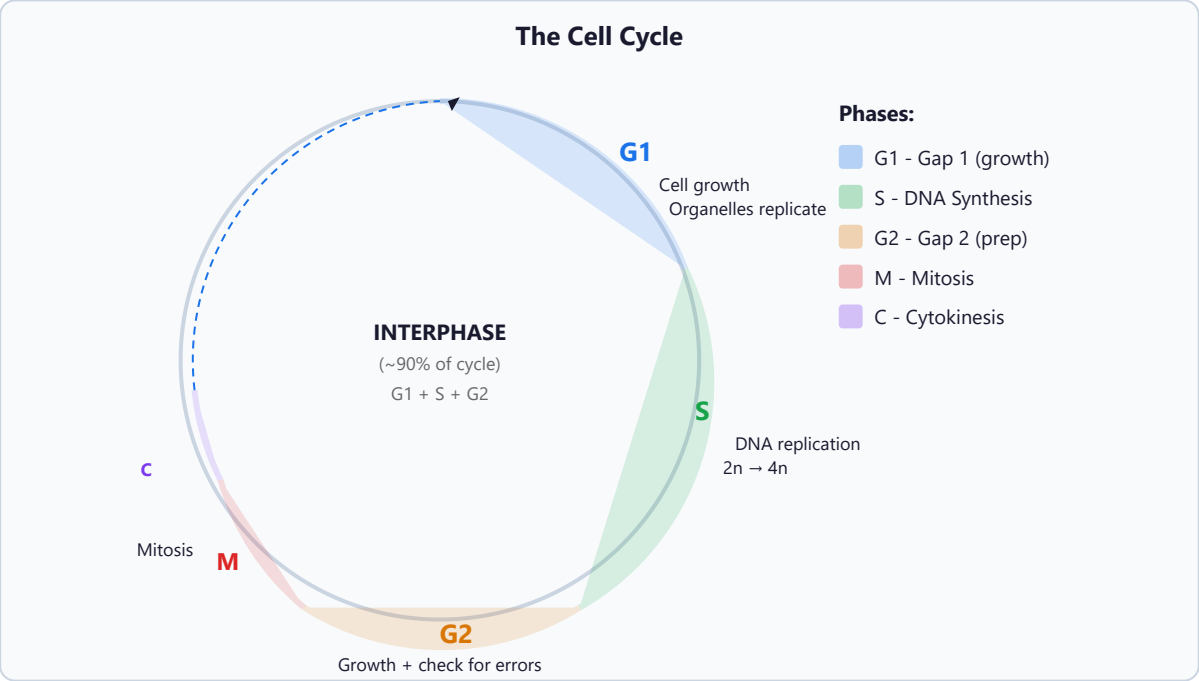
S3 — Voice of the Genome

3.1 Cell Division — Mitosis

The Cell Cycle

Cell cycle: The ordered sequence of events that takes place in a cell leading to its division into two daughter cells. It consists of interphase (G1, S, and G2 phases) and mitotic phase (mitosis + cytokinesis).



Interphase occupies the majority of the cell cycle (approximately 90% of the time) and is a period of intense metabolic activity, not a resting stage:

- **G1 (Gap 1):** Cell grows, organelles replicate, proteins are synthesised. The cell carries out its normal functions and increases in size.

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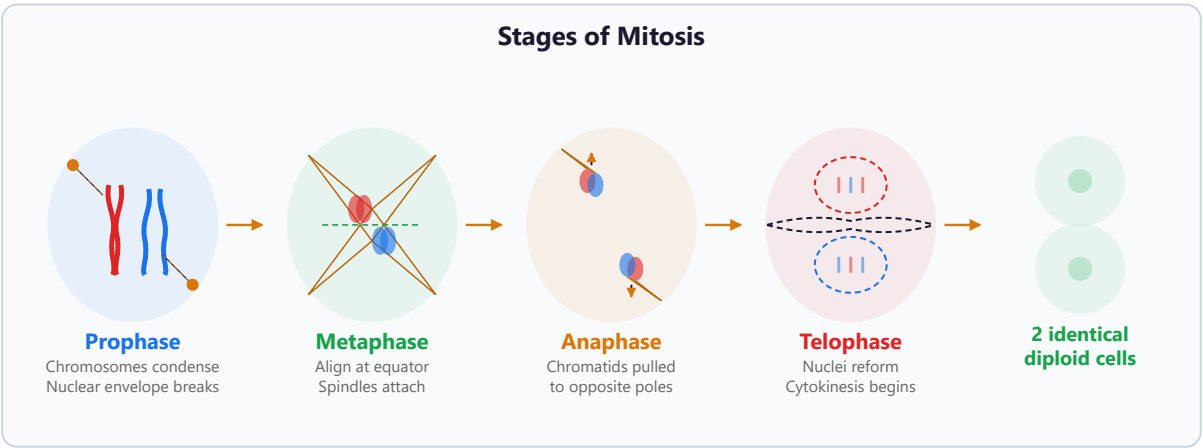
- **S (Synthesis):** DNA replication occurs. Each chromosome is duplicated to form two identical sister chromatids joined at the centromere. The cell's DNA content doubles from 2n to 4n.
- **G2 (Gap 2):** Further growth and protein synthesis. The cell checks for DNA replication errors and prepares for mitosis (e.g., ATP is produced, tubulin for spindle fibres is synthesised).

Stages of Mitosis

Stage	Key Events
Prophase	Chromosomes condense and become visible; each consists of two sister chromatids joined at the centromere. Centrioles migrate to opposite poles and spindle fibres begin to form. Nuclear envelope breaks down; nucleolus disappears.
Metaphase	Chromosomes align along the cell equator (metaphase plate). Spindle fibres attach to centromeres from both poles. This ensures each daughter cell receives one chromatid from each chromosome.
Anaphase	Centromeres divide. Spindle fibres shorten, pulling sister chromatids apart to opposite poles. Chromatids are pulled centromere-first, giving a characteristic V shape.
Telophase	Chromatids (now called chromosomes) reach the poles and decondense. Nuclear envelopes reform around each set of chromosomes. Nucleoli reappear.
Cytokinesis	The cytoplasm divides. In animal cells, a cleavage furrow pinches inward. In plant cells, a cell plate forms from the centre outward. Two genetically identical daughter cells are produced.



Mitosis produces two genetically identical diploid daughter cells from one parent cell. It is used for growth, repair of damaged tissue, replacement of worn-out cells (e.g., skin cells are replaced every 2-3 weeks, red blood cells every 120 days), and asexual reproduction (e.g., budding in yeast, vegetative propagation in plants). Maintaining the chromosome number is crucial for ensuring genetic stability across generations of cells.

Calculating the mitotic index: In practical work, the mitotic index measures the proportion of cells in a tissue that are undergoing mitosis. A high mitotic index indicates rapid cell division, which may indicate normal growth or, pathologically, a tumour.

$$\text{Mitotic index} = (\text{number of cells in mitosis} \,/\, \text{total number of cells observed}) \times 100\%$$

Cancer and uncontrolled cell division: Cancer results from mutations in genes that control the cell cycle, particularly **proto-oncogenes** (which normally promote cell division; when mutated into oncogenes, they are permanently activated) and **tumour suppressor genes** (which normally inhibit cell division or trigger apoptosis; when mutated, this brake is lost). For example, mutations in the p53 tumour suppressor gene are found in over 50% of human cancers. The p53 protein normally halts the cell cycle at G1/S checkpoint to allow DNA repair, or triggers apoptosis if damage is irreparable. Multiple mutations are typically required for cancer to develop (the multi-hit hypothesis).

Real-world application: Anti-cancer drugs such as vincristine and taxol (paclitaxel) target mitosis. Vincristine prevents spindle fibre formation by inhibiting tubulin polymerisation, so chromosomes cannot be separated during anaphase. Taxol stabilises microtubules so they cannot disassemble, also blocking cell division. Both drugs affect rapidly dividing cells, which is why they cause side effects in tissues with high mitotic rates (hair follicles, bone marrow, gut lining).

Exam Tip: Learn to identify stages of mitosis from microscope images. Prophase: visible chromosomes, no nuclear envelope. Metaphase: chromosomes lined up in the middle. Anaphase: chromatids being pulled apart. Telophase: two groups at poles, nuclear envelopes reforming.

3.2 Cell Division — Meiosis

Overview of Meiosis

Meiosis: A type of nuclear division that produces four genetically different haploid daughter cells from one diploid parent cell. It involves two successive divisions (meiosis I and meiosis II) and is essential for sexual reproduction and genetic variation.

Meiosis I is the **reduction division** (chromosome number halved from 2n to n). Key events in each stage:

- **Prophase I:** Homologous chromosomes pair up (synapsis), forming bivalents. Each bivalent consists of four chromatids (a tetrad). **Crossing over** occurs between non-sister chromatids at points called chiasmata, where segments of DNA are exchanged. This creates recombinant chromosomes with new allele combinations. The nuclear envelope breaks down and spindle fibres form.
- **Metaphase I:** Bivalents line up along the cell equator. The orientation of each bivalent is random (**independent assortment**), meaning that maternal and paternal chromosomes are distributed randomly to the two poles.
- **Anaphase I:** Homologous chromosomes (not sister chromatids) are pulled to opposite poles by spindle fibres. This is the step that halves the chromosome number.
- **Telophase I and cytokinesis:** Two haploid cells form, each containing one chromosome from each homologous pair (but each chromosome still consists of two sister chromatids).

Meiosis II resembles mitosis: centromeres divide and sister chromatids are separated to opposite poles. No further DNA replication occurs between meiosis I and II. Four haploid cells result, each genetically unique due to the combined effects of crossing over and independent assortment.

Significance of meiosis: Meiosis halves the chromosome number, ensuring that when two gametes fuse at fertilisation, the diploid number is restored (maintaining a constant chromosome number across generations). It also generates genetic variation, which is the raw material for natural selection and evolution.

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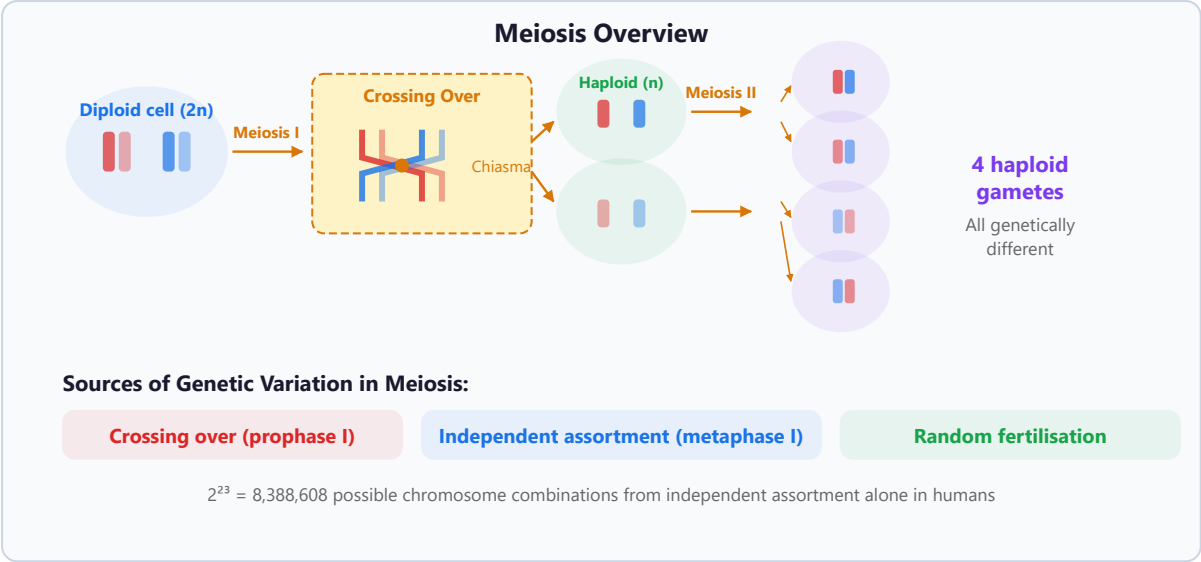

Sources of Genetic Variation

- **Independent assortment:** During metaphase I, the orientation of each bivalent is random. With 23 pairs of homologous chromosomes in humans, there are 2^{23} (over 8 million) possible combinations of maternal and paternal chromosomes in gametes.
- **Crossing over:** During prophase I, non-sister chromatids of homologous chromosomes exchange genetic material at chiasmata. This produces recombinant chromosomes with new combinations of alleles, further increasing genetic variation.
- **Random fertilisation:** Any male gamete can fuse with any female gamete, multiplying the possible genetic combinations enormously.
- **Mutation:** Random changes in DNA sequence create new alleles.

Comparing Mitosis and Meiosis

Feature	Mitosis	Meiosis
Number of divisions	One	Two
Daughter cells produced	Two	Four
Chromosome number	Maintained (diploid)	Halved (haploid)
Genetic composition	Identical to parent	Genetically different

Crossing over	Does not occur	Occurs in prophase I
Where it occurs	Somatic (body) cells	Reproductive organs (gonads)
Purpose	Growth, repair	Production of gametes

3.3 Cell Differentiation and Specialisation

Cell Differentiation

> **Differentiation:** The process by which an unspecialised cell becomes specialised for a particular function. This involves the expression of specific genes while other genes are switched off, leading to the production of specific proteins that give the cell its characteristic structure and function.

All cells in a multicellular organism contain the same DNA (genome), but different cells express different genes. Gene expression is controlled by transcription factors, epigenetic modifications (methylation, histone modification), and signalling molecules. Once a cell has fully differentiated, it typically cannot change into another cell type (though there are exceptions, such as in regeneration).

Examples of Specialised Cells

Cell Type	Specialisation	Function
Red blood cell (erythrocyte)	No nucleus (more space for haemoglobin); biconcave disc shape (large SA:V ratio); flexible	Transport oxygen
Neutrophil	Lobed nucleus; many lysosomes; flexible membrane	Engulf and destroy pathogens (phagocytosis)
Sperm cell	Streamlined shape; flagellum; many mitochondria; acrosome with digestive enzymes	Swim to and fertilise egg
Root hair cell	Long extension (hair) increasing surface area; many mitochondria for active transport	Absorb water and mineral ions from soil

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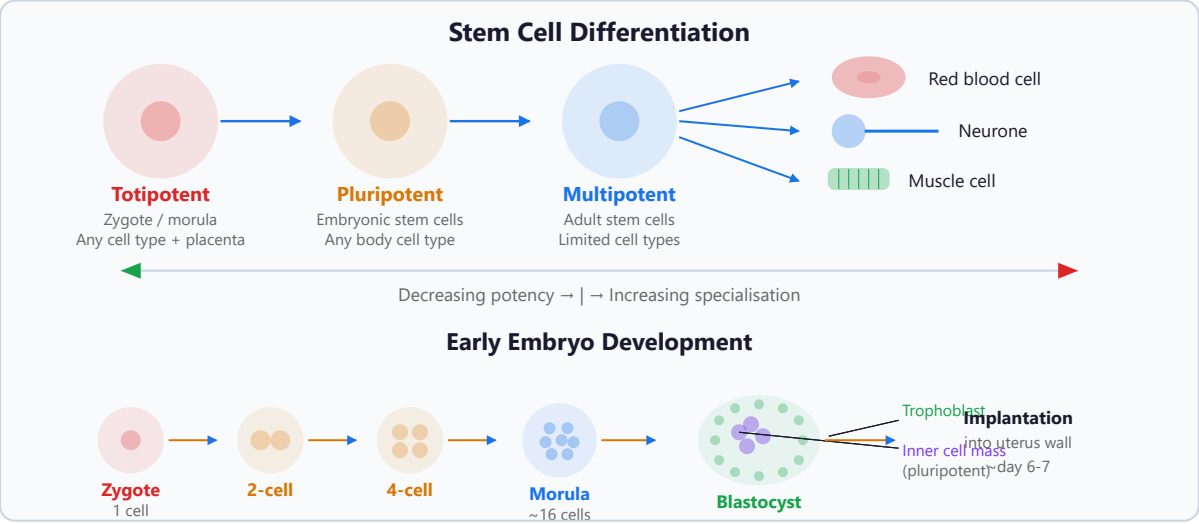
Palisade mesophyll cell	Many chloroplasts; positioned at top of leaf; elongated shape	Photosynthesis

3.4 Stem Cells and Development

Types of Stem Cells

Stem cell: An undifferentiated cell capable of self-renewal (dividing to produce more stem cells) and differentiation into one or more specialised cell types.

- **Totipotent stem cells:** Can differentiate into any cell type in the body plus extraembryonic tissues (e.g., placenta). Found in the zygote and cells up to the 8-cell (morula) stage.
- **Pluripotent stem cells:** Can differentiate into any cell type of the three germ layers (ectoderm, mesoderm, endoderm) but not extraembryonic tissues. Found in the inner cell mass of the blastocyst (embryonic stem cells).
- **Multipotent stem cells:** Can differentiate into a limited range of related cell types. Found in adults (e.g., haematopoietic stem cells in bone marrow produce all blood cell types).
- **Unipotent stem cells:** Can only produce one cell type but retain self-renewal capacity. Example: skin stem cells producing keratinocytes.



Induced Pluripotent Stem Cells (iPSCs)

Induced pluripotent stem cells (iPSCs) are adult somatic cells (e.g., skin fibroblasts) that have been reprogrammed to an embryonic-like pluripotent state by introducing specific transcription factors (originally the Yamanaka factors: Oct4, Sox2, Klf4, c-Myc, discovered by Shinya Yamanaka in 2006, earning him the Nobel Prize in 2012). iPSCs can differentiate into almost any cell type and avoid the ethical issues associated with embryonic stem cells, since no embryo is destroyed. They also offer the possibility of patient-specific therapies (using the patient's own cells reduces immune rejection). However, concerns remain about potential tumour formation (especially from the c-Myc oncogene), incomplete reprogramming, epigenetic abnormalities, and low efficiency of the reprogramming process.

Applications of stem cell research:

- **Regenerative medicine:** Growing replacement tissues (e.g., skin grafts for burn victims, retinal cells for macular degeneration, dopamine-producing neurons for Parkinson's disease, insulin-producing beta cells for type 1 diabetes).
- **Drug testing:** Testing new drugs on human cells differentiated from stem cells, reducing the need for animal testing and providing more relevant results.
- **Disease modelling:** Creating iPSCs from patients with genetic diseases to study disease mechanisms in the laboratory.

Ethical Issues

Embryonic stem cell research raises ethical concerns because extracting cells from the inner cell mass destroys the embryo. Arguments in favour include the potential to treat serious diseases (Parkinson's, diabetes, spinal cord injuries). Arguments against include the view that the embryo has moral status from conception, and that alternative approaches (adult stem cells, iPSCs) may achieve similar goals without embryo destruction. Legislation varies between countries.

3.5 Gene Expression and Development

Control of Gene Expression

Although every cell contains the full genome, only certain genes are expressed in each cell type. Gene expression is regulated at multiple levels:

- **Transcriptional control:** Transcription factors bind to promoter/enhancer regions to activate or repress transcription. This is the most important level of control.

- **Epigenetic control:** DNA methylation (adding methyl groups to cytosine bases) typically silences genes. Histone modification (acetylation, methylation) affects how tightly DNA is packaged and thus whether genes are accessible for transcription.
- **Post-transcriptional control:** Alternative splicing of mRNA can produce different proteins from the same gene. mRNA stability and degradation rates also affect how much protein is produced.
- **Translational and post-translational control:** Regulation of translation rate; protein modification (phosphorylation, glycosylation); protein degradation.

Development and Body Plans

During embryonic development, the body plan is established through complex patterns of gene expression controlled by master regulator genes. Homeobox genes (including Hox genes) encode transcription factors that specify the identity of body segments along the anterior-posterior axis. Mutations in Hox genes can cause dramatic developmental abnormalities, such as a leg growing where an antenna should be in Drosophila.

Cell signalling is crucial for coordinating development. Cells communicate through chemical signals (morphogens, growth factors, hormones) that bind to receptors and trigger intracellular signalling cascades, ultimately affecting gene expression. The concentration gradient of morphogens helps cells determine their position and differentiate accordingly (positional information).

Exam Tip: Remember that differentiation is about which genes are switched on or off, not about changes to DNA sequence. All body cells have the same genome (with a few exceptions like mature red blood cells and lymphocytes). The key concept is differential gene expression.

Programmed Cell Death (Apoptosis)

Apoptosis: Programmed cell death — a controlled, orderly process in which cells self-destruct without causing inflammation. The cell shrinks, chromatin condenses, DNA is fragmented, and the cell breaks into apoptotic bodies that are phagocytosed by neighbouring cells.

Apoptosis is essential during development (e.g., separating digits, eliminating excess neurons, removing webbing between fingers) and throughout life (e.g., destroying damaged or potentially cancerous cells, removing cells infected by viruses, maintaining tissue homeostasis). It is controlled by genes such as p53

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(tumour suppressor) and the Bcl-2 family. Failure of apoptosis can lead to cancer; excessive apoptosis can lead to degenerative diseases.

Exam Tip: Distinguish between apoptosis (controlled, no inflammation, beneficial) and necrosis (uncontrolled cell death due to injury, causes inflammation, harmful). Apoptosis requires ATP and gene expression; necrosis does not.

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